

mdstats FINAL-GPU1 Workstation Qualification Runbook

mdstats development

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CURRENT TARGET-SIZE-v5 HANDOFF

This runbook describes the current one-shot FINAL-GPU1 handoff for the target-size-v5 software generation. It is intentionally independent of the retired SIZE-FIDELITY2/MVMIGRATE1 activation workflow. Target-size v5 is already the production architecture; FINAL-GPU1 qualifies accelerator execution and release performance only.

1. Inputs

Prepare one workstation directory containing:

- the exact release source archive being qualified;
- the locked mace-mh-1.model and mace-mpa-0-medium.model files;
- the target/replay data required by the release-matched campaign;
- the project offline dependency bundle; and
- a writable qualification root.

Do not replace either foundation model. The tool verifies both model SHA-256 identities against the locked mdstats constants.

2. Runtime prerequisite

Activate the CUDA environment intended to authorize the release. It must contain the release-matched MACE/e3nn stack plus the CuEquivariance core, Torch frontend, and CUDA operations package appropriate to the installed PyTorch/CUDA runtime.

Missing accelerator capability is a failed/deferred runtime, not permission to alter scientific settings or silently fall back inside the same qualification record.

3. Verify and unpack the release

Verify the source archive with the checksum manifest supplied with the final bundle, then unpack it without editing source. The exact archive SHA-256 is part of every handoff registration.

Example:

```
sha256sum -c VERIFY_BUNDLE_SHA256.txt
mkdir -p work/source
unzip -q <release-archive.zip> -d work/source
cd work/source/<release-directory>
```

4. Preflight

```
python tools/run_mlff_final_gpu_qualification.py preflight \
--mh1-model ../../inputs/mace-mh-1.model \
--mpa0-model ../../inputs/mace-mpa-0-medium.model \
--release-archive ../../<release-archive.zip> \
--output ../../final-gpu1/preflight.json
```

The current preflight is the target-size-v5 v11 contract. A ready_for_final_gpu_execution result requires the locked models, a readable release archive, CUDA availability, and a positive CUEQ-DEP1 runtime freeze.

5. Initialize one immutable handoff root

```
python tools/run_mlff_final_gpu_qualification.py init \  
  --root ../../../../final-gpu1/run-001 \  
  --mh1-model ../../../../inputs/mace-mh-1.model \  
  --mpa0-model ../../../../inputs/mace-mpa-0-medium.model \  
  --release-archive ../../../../<release-archive.zip>
```

The current policy creates a 16-item matrix: 8 must_pass, 6 measure_only, and 2 optional gates. Do not re-run init on a populated root. Replacement evidence belongs in a new run root.

6. Capture CUEQ-DEP1 once

```
mkdir -p ../../../../final-gpu1/run-001/raw  
python tools/capture_mlff_cueq_depl_runtime.py \  
  --supplied-artifact ../../../../inputs/mace-mh-1.model \  
  --supplied-artifact ../../../../inputs/mace-mpa-0-medium.model \  
  --output ../../../../final-gpu1/run-001/raw/cueq_depl_runtime.json  
  
python tools/run_mlff_final_gpu_qualification.py record \  
  --root ../../../../final-gpu1/run-001 \  
  --gate CUEQ_DEP1_RUNTIME_FREEZE \  
  --evidence ../../../../final-gpu1/run-001/raw/cueq_depl_runtime.json \  
  --disposition auto
```

All runtime-bound evidence must authenticate to this exact runtime digest.

7. Execute the current matrix

7.1 Must-pass gates

Gate ID	Required evidence
CUEQ_DEP1_RUNTIME_FREEZE	positive release-authorizing CuEq/CUDA runtime freeze
E3NN_BASELINE_COMPLETE_CAMPAIGN	complete optimized e3nn production-representative baseline
SIZE_FIDELITY1_EXHAUSTIVE_CALIBRATION	release qualification of the current size-screen calibration policy
PERF_P2R_WHOLE_FUNNEL_GPU_PERFORMANCE	whole-funnel GPU/control-plane performance with restart/decision equivalence
VRAM1_PERF_P4_ACCELERATOR_MEMORY_THROUGHPUT	VRAM/headroom/OOM/backoff and bounded-pipeline throughput evidence
CUEQ_PHASE1_TRAINING_ONLY_QUALIFICATION	paired e3nn/CuEq training qualification on one frozen runtime
REPLAY_UNIFY1_GPU_PSEUDOLABEL_EXECUTION	release-authoritative replay pseudolabel execution and cache/restart evidence
PERF_CERT1_END_TO_END_CERTIFICATION	end-to-end certification against the authoritative e3nn baseline

7.2 Measure-only gates

Register evidence for:

- PREC3_REAL_CUEQ_ACTIVATION
- MH1_ACCEL1_CUEQ_NUMERICAL_PARITY
- MH1_DATA6_1_CUEQ_DESCRIPTOR_SELECTION_PARITY
- MH1_TRAIN1_CUEQ_TRAINING_REALIZATION
- MH1_CERT1_GENERATED_DEFAULT_CUEQ_MATRIX
- PERF_P5_ACCELERATOR_PERSISTENCE_REUSE

These measurements must be complete even if a particular optimization remains disabled.

7.3 Optional gates

- CUEQ_PHASE2_SELECTED_HEAD_SOURCE_EXECUTION_OPTIONAL
- MH1_DEPLOY1_MLIAP_EXPORT_AND_LAMMPS_RUN0

8. Run the scientific campaign normally

Use the normal staged CLI. Do not bypass target-size selection or held-out boundaries:

```
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> doctor
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> prepare
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> preflight
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> train
```

```
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> evaluate
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> verify
```

The campaign itself owns the fixed-eight target-size population and configurable fidelity path:

```
REPAIR2 -> MVQUAL2 -> Q -> coarse(n1) -> short(n2) -> final-screen(n3) -> full horizon(n)
        -> selected_target_size -> selected REPAIR2 prefix
        -> held-out CV/EVAL/VERIFY
```

FINAL-GPU1 must not create rescue sizes, migrate old target ladders, or alter a selected target size.

9. Register evidence

For every matrix item, use the handoff tool rather than editing the manifest:

```
python tools/run_mlff_final_gpu_qualification.py record \
--root ../../../../final-gpu1/run-001 \
--gate <GATE_ID> \
--evidence ../../../../final-gpu1/run-001/raw/<evidence.json> \
--disposition auto
```

The registration binds the evidence bytes, schema/content digest where available, release archive, and CuEq runtime digest where required.

10. Verify integrity before reduction

```
python tools/run_mlff_final_gpu_qualification.py verify \
--root ../../../../final-gpu1/run-001
```

Any source/model/evidence mutation, policy/matrix drift, path escape, missing runtime binding, or inconsistent producer status fails closed.

11. Reduce FINAL-GPU1

```
python tools/run_mlff_final_gpu_qualification.py reduce \
--root ../../../../final-gpu1/run-001 \
--runtime ../../../../final-gpu1/run-001/raw/cueq_depl_runtime.json \
--perf-cert1 ../../../../final-gpu1/run-001/raw/perf_cert1_qualification.json \
--output ../../../../final-gpu1/run-001/FINAL_GPU1_QUALIFICATION.json
```

A positive record requires every must-pass gate to pass, every measure-only gate to be complete, handoff integrity to pass, and all runtime/release/model identities to agree.

A pass may recommend an accelerator profile but does not directly change generated defaults. The handoff record keeps generated-default authorization false.

12. Retired target-size migration workflow

Do **not** run SIZE-FIDELITY2, MVMIGRATE1 learning controls, or a target-data migration activation command. Those belonged to historical campaign generations and are not prerequisites for target-size v5. Current campaigns reject obsolete derived migration/rescue state and rebuild the target-size authority from authenticated REPAIR2/MVQUAL2 state.